Two new questions on the same topic as the exam question, a little more open-ended. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

The golden ratio is the number $\varphi = \dfrac{1+\sqrt{5}}{2}$.

(a) Show that $\varphi^2 = \varphi + 1$. **[3]**

(b) Hence show that $\dfrac{1}{\varphi} = \varphi - 1$. **[2]**

(c) Show that $\varphi^3 = 2 + \sqrt{5}$. **[3]**

2 Given that $(p+q\sqrt{3})(2-\sqrt{3}) = 1+\sqrt{3}$, where p and q are integers, find the values of p and q. **[4]**

Question 1

(a) $\varphi^2 = \dfrac{1+2\sqrt{5}+5}{4} = \dfrac{3+\sqrt{5}}{2}$ and $\varphi + 1 = \dfrac{1+\sqrt{5}}{2} + 1 = \dfrac{3+\sqrt{5}}{2}$

(b) Divide $\varphi^2 = \varphi + 1$ by φ: $\varphi = 1 + \dfrac{1}{\varphi}$, so $\dfrac{1}{\varphi} = \varphi - 1$

(c) $\varphi^3 = \varphi \cdot \varphi^2 = \varphi(\varphi+1) = \varphi^2 + \varphi = 2\varphi + 1 = (1+\sqrt{5}) + 1 = 2 + \sqrt{5}$

Question 2

Expanding: $(2p-3q) + (2q-p)\sqrt{3} = 1 + \sqrt{3}$. Equating rational and irrational parts: $2p - 3q = 1$ and $-p + 2q = 1$. Solving gives $q = 3$, $p = 5$. (Check: $(5+3\sqrt{3})(2-\sqrt{3}) = 10 - 5\sqrt{3} + 6\sqrt{3} - 9 = 1 + \sqrt{3}$.)